TEAM UNOSYN · EGYPTIAN CHINESE UNIVERSITY

# VitalOrbit

An autonomous onboard health system for deep-space crews.

When Earth is **19 light-minutes** away, the crew is its own first responder.



THE PROBLEM

# On Mars, help can't be 40 minutes away.

Today, astronaut health depends on flight surgeons on the ground. On a Mars mission that link breaks down.

**4–24 min**

**one-way signal delay**

A question and its answer take up to 48 minutes; in our mission model Earth is 19.1 light-minutes away on day 432.

**~2 weeks**

**solar conjunction blackout**

When the Sun sits between Mars and Earth, there is no link at all.

**5 hazards**

**NASA's RIDGE risks**

Radiation, Isolation, Distance, Gravity (bone, muscle, vision), hostile Environment: slow damage that no one on Earth sees in time.

THE SOLUTION

# One system that detects, guides and remembers.

### Personal anomaly detection

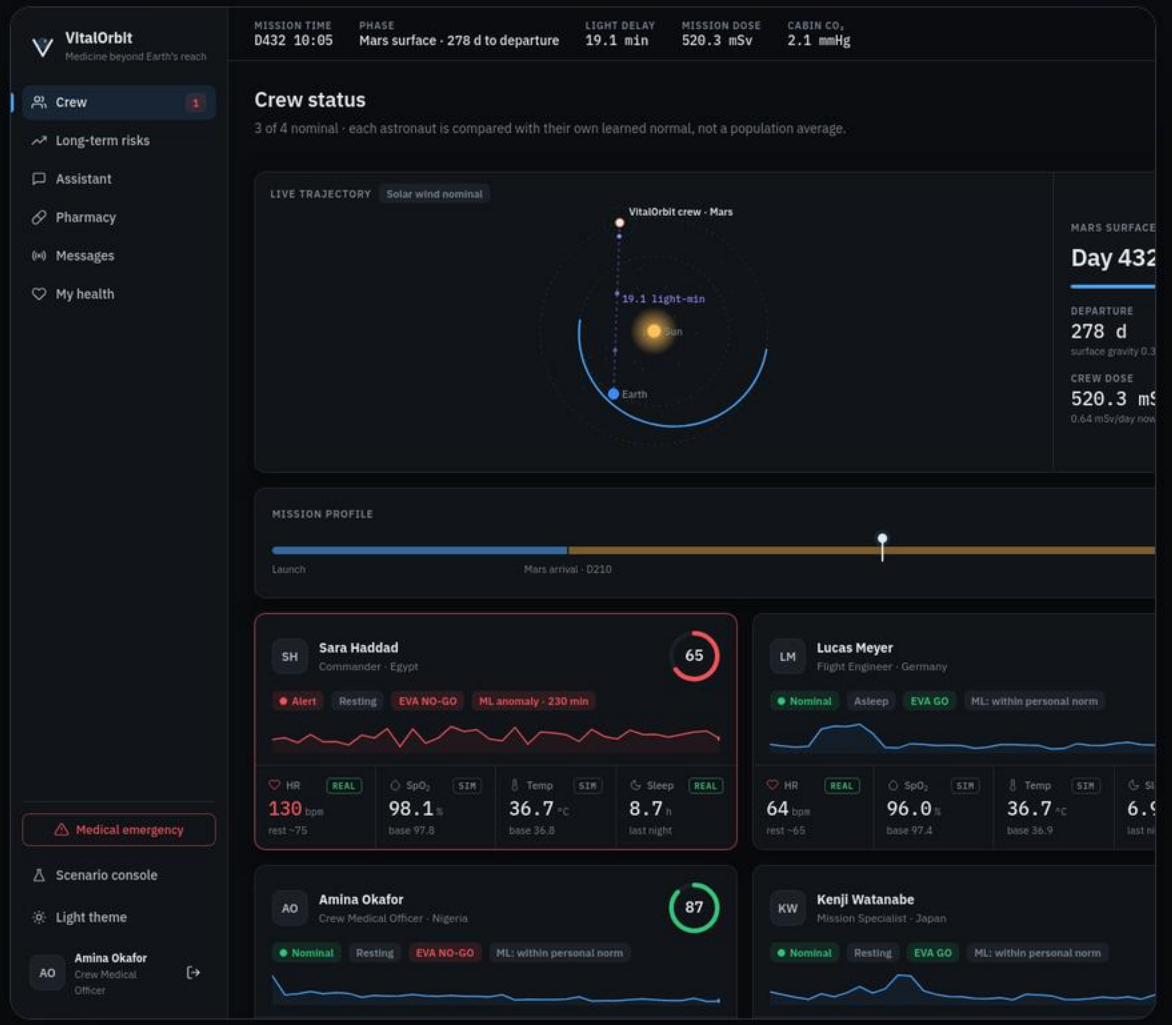
An ML model per astronaut, trained on their own real wearable data, plus clinical safety limits.

### Long-mission risk forecasting

Radiation vs career limit, bone and muscle loss, vision (SANS), mood: projected to mission end.

### Offline medical assistant + smart pharmacy

Guided first aid with timers, voice and Arabic; every drug checked for stock and radiation-shortened expiry.



HOW IT WORKS

# From a heartbeat to a decision, onboard.

REAL SIM

### Vitals

Fitbit heart rate, activity, sleep (public dataset); simulated $SpO_2$ and temperature.

→

### Personal ML

Isolation Forest on deviations from each person's own rhythm; alert after 30 min of persistence.

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### Check-in

"How do you feel?" The answer labels the alert and opens the right protocol.

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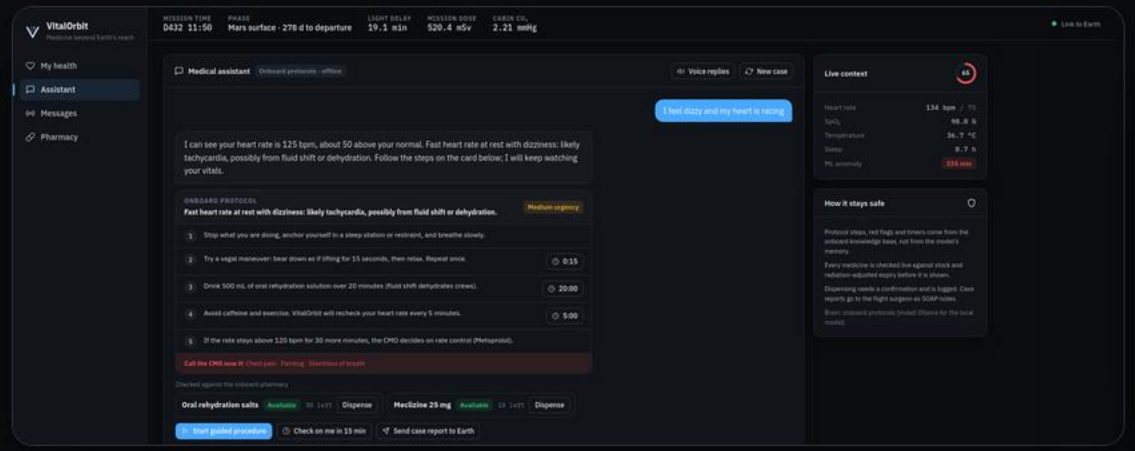
### Assistant + pharmacy

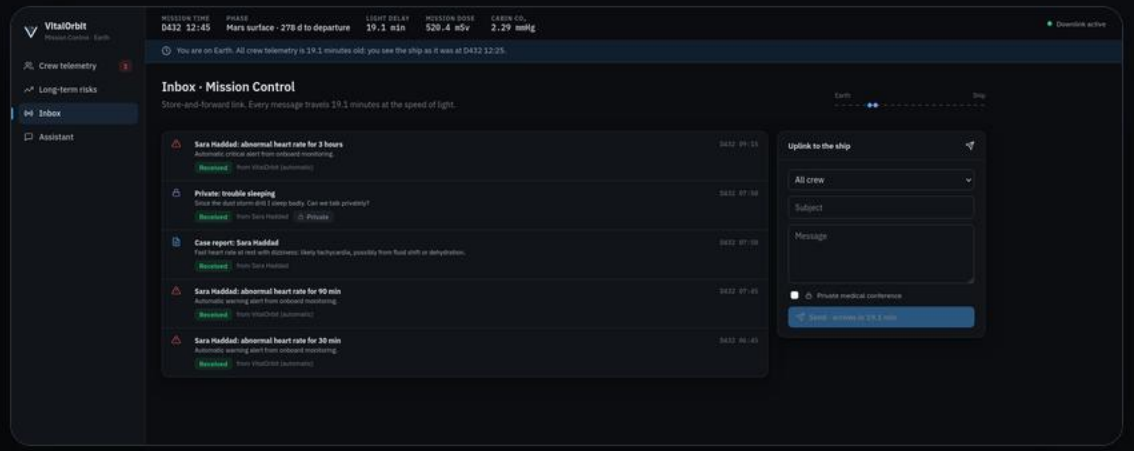
Steps, timers, red flags; medicine checked live. Dispensing is logged.

→

### Earth, delayed

SOAP case report and alerts arrive 19 min later; private line to the flight surgeon.



DOES IT WORK?

# Measured on real wearable data.

3,456 real readings from 4 people, plus 12 injected emergencies (tachycardia, low oxygen, fever).

**12 / 12**
emergencies detected, correct vital named every time

**45 min**
median time to the first alert, critical within 3 h in 100 %

**0.08**
false warnings per crew-day (1 in 12 crew-days)

**0**
false critical alerts

## Honest by design

Every number in the app is tagged REAL, SIM or MODEL.

- Fitbit dataset, Zenodo 53894 (CC-BY 4.0)
- Radiation: Curiosity RAD, 1.8 mSv/day cruise, 0.64 on Mars
- Career limit 600 mSv (NASA-STD-3001)
- Bone loss and countermeasures: LeBlanc 2000, Smith 2012
- Drug potency in flight: Du et al. 2011
- Light delay and conjunction from an orbital model

IMPACT

# Built for Mars. Useful wherever the doctor is far away.

### For crews

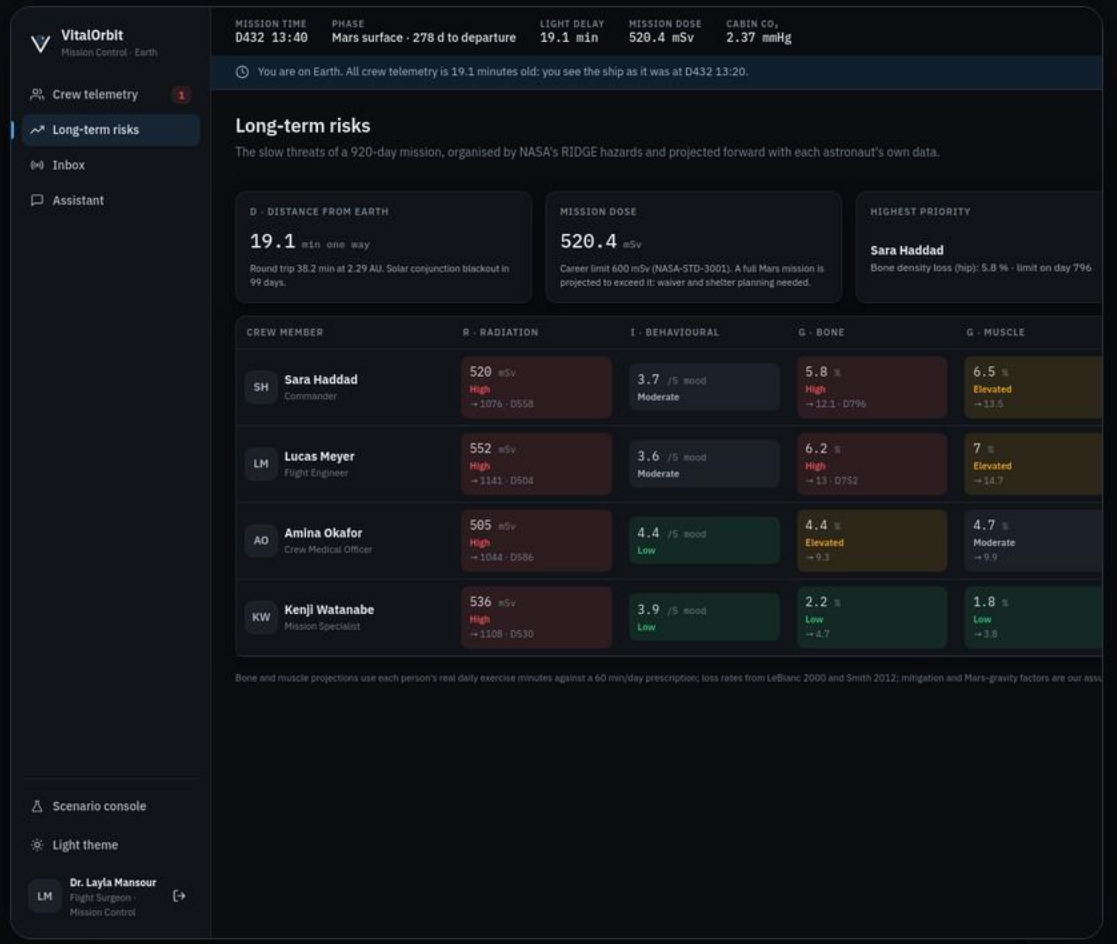
Earlier warnings, guided care in minutes instead of a 40-minute round trip, privacy for each astronaut.

### For flight surgeons

Structured SOAP reports and long-term risk trends, ready when the link returns.

### On Earth

Antarctic stations, submarines, offshore rigs, remote villages and field hospitals.



THANK YOU

# Medicine beyond Earth's reach.

Code and run instructions
github.com/3AMURR/VitalOrbit

Run it: `./run.sh` · self-running demo: `localhost:8000/?tour=1`

Team UnoSyn · Egyptian Chinese University · Built with help from Claude (Anthropic)

